

DL Graded Quiz M2

Question 1

Once an MLP is able to learn a feature from a location in an image, it will detect it only in that location. In the context of processing candlestick charts to predict stock returns with MLPs, which of the following is **true**?

- ☐ This is an advantage because the network may learn irrelevant features for prediction.
- ☐ This is an advantage because the same candlestick price pattern at different points in time may indicate different directions for future prices.
- ☐ This is a disadvantage because the network will generalize well.
- ☒ This is a disadvantage because the network will identify patterns that depend on their location.

Question 2

Consider the following input image with two channels:

$$\begin{pmatrix} 1 & 2 & 4 \\ 3 & 7 & 2 \\ 2 & 0 & 2 \end{pmatrix}$$
$$\begin{pmatrix} 0 & 3 & 2 \\ 5 & 4 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

Determine which of the following is the outcome of a 2x2 max-pooling layer.

☐ $\begin{pmatrix} 7 & 7 \\ 7 & 5 \end{pmatrix}$

$\begin{pmatrix} 3 & 4 \\ 3 & 4 \end{pmatrix}$

☐ $\begin{pmatrix} 4 & 4 \\ 7 & 5 \end{pmatrix}$

$\begin{pmatrix} 5 & 4 \\ 5 & 4 \end{pmatrix}$

☐ $\begin{pmatrix} 7 & 7 \\ 5 & 5 \end{pmatrix}$

$\begin{pmatrix} 5 & 4 \\ 5 & 4 \end{pmatrix}$

☒ $\begin{pmatrix} 7 & 7 \\ 7 & 7 \end{pmatrix}$

$\begin{pmatrix} 5 & 4 \\ 5 & 4 \end{pmatrix}$

Question 3

Consider the following input image:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

Consider the following kernel:

$$\begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix}$$

Determine which of the following is the outcome of the convolution operator after padding 2 rows and 2 columns.

☐ $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 4 & 3 & 5 & 2 \\ 4 & 5 & 5 & 2 \\ 3 & 1 & 3 & 1 \end{pmatrix}$

☐ $\begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix}$

☐ 5

☒ $\begin{pmatrix} 1 & 2 & 2 & 1 \\ 4 & 5 & 5 & 2 \\ 4 & 2 & 4 & 2 \\ 3 & 1 & 3 & 1 \end{pmatrix}$

Question 4

Determine which of the following combinations of elements are trained in a basic convolutional neural network.

- ☐ The pad and stride parameters
- ☐ The bias terms, the elements of each convolutional kernel, plus the number of layers in the network
- ☐ The bias terms and the elements of each convolutional kernel, plus the pad and the stride parameters
- ☒ The bias terms and elements of each convolutional kernel

Question 5

Consider a CNN composed of three convolutional layers, each with 3×3 kernels and a stride of 2, and padding is chosen so that the output layers have the same

dimensionality as the inputs. The lowest layer outputs 50 channels, the middle one outputs 200, and the top one outputs 400. The input images are RGB images of 200×300 pixels. What is the total number of parameters in the CNN?

- ☐ 131,584
- ☐ 1,025,300
- ☐ 812,000
- ☒ 720,400

Question 6

Adding a pooling layer to a convolutional neural network does which of the following?

- ☐ It increases the number of trainable parameters of the model by an amount equal to the number of output channels.
- ☐ It increases the number of trainable parameters of the model by an amount equal to the shape of the input.
- ☐ It does not increase the number of trainable parameters of the model
- ☒ It increases the number of trainable parameters of the model by an amount equal to the shape of the kernel.

Question 7

Consider the following input image:

$$\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

Consider the following kernel:

$$\begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix}$$

Determine which of the following is the outcome of the convolution operator after padding 2 rows and 2 columns.

☐ 5

☐ $\begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix}$

☐ $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 4 & 3 & 5 & 2 \\ 4 & 5 & 5 & 2 \\ 3 & 1 & 3 & 1 \end{pmatrix}$

☒ $\begin{pmatrix} 1 & 2 & 2 & 1 \\ 4 & 5 & 5 & 2 \\ 4 & 2 & 4 & 2 \\ 3 & 1 & 3 & 1 \end{pmatrix}$

Question 8

Consider a convolutional layer with input shape 25x25 and three channels. We apply a kernel with shape 5x5, zero padding, and a stride of 2 horizontally and vertically. The number of output channels is one. The shape of each channel of the output layer is equal to which of the following?

☐ 15x15

☐ 11x11

☐ 2x2

Question 9

Consider the candlestick chart application of Lesson 4. If a CNN is able to obtain a good accuracy in the training set and in validation, then what does this mean?

- ☐ This means that the network overfits the data.
- ☐ This means that the features learned by the network are repeatedly associated with a certain direction in future stock returns.
- ☐ This means that the features learned by the network are not repeatedly associated with a certain direction in future stock returns.
- ☒ This means that the network underfits the data.

Question 10

In a layer of a neural network that includes batch normalization, what do we do?

- ☐ We only add a shift parameter to train in the model.
- ☐ We scale the input so that it has mean zero and unit standard deviation in the current mini-batch.

- ☐ We scale the input so that it has mean zero and unit standard deviation in the whole training sample.
- ☒ We do not add trainable parameters to the model.

Question 11

Consider the following input image:

$$\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

Consider the following kernel:

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

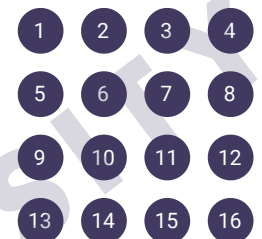
Determine which of the following is the outcome of the convolution operator.

- ☐ $\begin{pmatrix} 4 & 4 \\ 3 & 4 \end{pmatrix}$
- ☐ $\begin{pmatrix} 3 & 4 \\ 2 & 2 \end{pmatrix}$
- ☐ $\begin{pmatrix} 3 & 4 \\ 1 & 3 \end{pmatrix}$
- ☒ $\begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix}$



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QUESTIONS



Question 12

In a convolutional neural network with one convolutional layer, kernel shape 2x2, two input channels, one output channel, and stride of 2

vertically and horizontally, what is the number of trainable parameters?

- ☐ 11
- ☐ 9
- ☐ 10
- ☒ 12

Question 13

Consider the candlestick chart application of Lesson 4. Which of the following is **true** about using the grayscale version of the images as an input to a CNN?

- ☐ We are **NOT** providing precise information about the trading volume during each day that appears in the chart.
- ☐ We are **NOT** providing precise information about the sign of the stock return during each day that appears in the chart.
- ☐ We are **NOT** providing precise information about the range of stock prices during each day that appears in the chart.
- ☒ It is irrelevant because CNNs do not consider the color dimension.

Question 14

Consider the following input images from a mini-batch of

two examples in a convolutional neural network:

$$\begin{pmatrix} 1 & 2 & 4 \\ 3 & 7 & 2 \\ 2 & 0 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 5 & 4 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

Determine which of the following is the outcome of a batch normalization layer, before applying the activation function, that applies a shift parameter of 0.2 and a scale parameter of 0.75.

☐ $\begin{pmatrix} 0.11 & -0.72 & 0.11 \\ -0.30 & 0.11 & 0.94 \\ 0.52 & 2.18 & 0.11 \end{pmatrix}$

$$\begin{pmatrix} -0.31 & -0.72 & -0.31 \\ -0.72 & 0.52 & 0.11 \\ 1.35 & 0.94 & -0.31 \end{pmatrix}$$

☐ $\begin{pmatrix} 0.52 & 2.18 & 0.11 \\ 0.11 & -0.72 & 0.11 \\ -0.30 & 0.11 & 0.94 \end{pmatrix}$

$$\begin{pmatrix} 1.35 & 0.94 & -0.31 \\ -0.31 & -0.72 & -0.31 \\ -0.72 & 0.52 & 0.11 \end{pmatrix}$$

☐ $\begin{pmatrix} 0.52 & 2.18 & 0.11 \\ -0.30 & 0.11 & 0.94 \\ 0.11 & -0.72 & 0.11 \end{pmatrix}$

$$\begin{pmatrix} 1.35 & 0.94 & -0.31 \\ -0.72 & 0.52 & 0.11 \\ -0.31 & -0.72 & -0.31 \end{pmatrix}$$

☒ $\begin{pmatrix} -0.30 & 0.11 & 0.94 \\ 0.52 & 2.18 & 0.11 \\ 0.11 & -0.72 & 0.11 \end{pmatrix}$

$$\begin{pmatrix} -0.72 & 0.52 & 0.11 \\ 1.35 & 0.94 & -0.31 \\ -0.31 & -0.72 & -0.31 \end{pmatrix}$$

Question 15

In the context of convolutional neural networks, stride allows to do which of the following?

- ☐ Reduce the dimensions of the kernel
- ☐ Reduce the number of trainable parameters in the model
- ☐ Reduce the computational costs of convolutional layers
- ☒ Reduce the number of output channels of a layer

Question 16

Consider a convolutional layer with input shape 25×25 and two channels. We apply a kernel with shape 5×5 , a padding of 3 on all sides, and a stride of 1. The number of output channels is two. The shape of each channel of the output layer is equal to which of the following?

- ☐ 27×27
- ☐ 10×10
- ☐ 5×5
- ☒ 25×25

SUBMIT